

SYNOPSIS

AI-Powered Smart Digital Attendance System Using Face Recognition

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Project Title: Smart Digital Attendance System

Domain: Artificial Intelligence / Computer Vision / Web Application

Platform: Python (Flask), InsightFace, PostgreSQL

1. Introduction

Attendance management is a basic but critical activity in colleges and universities. Traditional methods such as paper registers and manual roll calls are slow, error-prone, and easy to manipulate. Even barcode or RFID systems need extra hardware and still depend on student honesty.

This project proposes an **AI-powered Smart Digital Attendance System** that uses **face recognition** to mark attendance automatically. Teachers can enroll students with guided face capture, train face embeddings, and mark attendance in two modes:

1. **Live camera mode** (desk / entry style)
2. **Classroom group photo mode** (many students in one image)

The system is designed around real college workflow: **classes, subjects, teacher assignments, role-based access, attendance uniqueness per day, PostgreSQL storage, and an AI Attendance Copilot** for quick academic queries.

2. Problem Statement

Colleges face these problems with conventional attendance:

- Manual marking wastes teaching time
- Proxy attendance is common
- Records are hard to search, analyze, and export
- Large classes make roll call impractical
- Existing face apps often ignore college structure (class/subject/teacher rights)

There is a need for a system that is:

- Accurate enough for classroom use

- Structured for college administration
 - Scalable toward thousands of students
 - Transparent, with teacher control over weak matches
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3. Objectives

1. To develop a web-based digital attendance system using face recognition.
 2. To enroll students with multiple face samples and store compact face embeddings.
 3. To support live recognition and classroom group-photo recognition.
 4. To enforce college rules: class/subject mapping, teacher access control, and one present mark per student per class–subject–day.
 5. To improve reliability using a **Review workflow** for weak matches.
 6. To store data permanently in **PostgreSQL** for concurrent college use.
 7. To provide an **AI Attendance Copilot** for questions like absences and low-attendance alerts.
 8. To prepare the system for future college portal integration.
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4. Scope of the Project

In scope

- Admin: create classes, subjects, teachers, assignments
- Teacher: add/manage students, capture faces, train model, mark attendance
- Student: view own attendance (when login is created)
- Face recognition using InsightFace (buffalo_1, ArcFace-family embeddings)
- Class-scoped matching (compare only within selected class)
- Records, analytics, CSV export
- Classroom photo review before final save
- Rules-based AI Copilot
- PostgreSQL as permanent database

Out of scope (future)

- Full college ERP / fee / exam modules
 - Mobile native apps
 - SSO / national ID integration
 - Fully offline neural training from scratch
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5. Proposed System Overview

The system follows a clear pipeline:

```
Student enrollment (details + face capture)
    ↓
Train Model → face embeddings / centroids stored in DB
    ↓
Teacher selects Class + Subject
    ↓
Live camera OR Classroom photo
    ↓
AI face detect → embed → class-scoped match
    ↓
Review weak matches (teacher confirmation)
    ↓
Save attendance (unique per day) in PostgreSQL
    ↓
Records / Analytics / AI Copilot
```

Important design choice: the project does **not** train a deep neural network from scratch. It uses a pretrained InsightFace model to generate embeddings, then matches students using cosine similarity against per-student centroids. This is practical, fast, and suitable for a college project with strong technical justification.

6. Methodology

6.1 Face Recognition Method

- Detection and embedding: InsightFace buffalo_1
- Representation: 512-D face embedding
- Gallery: per-student centroid (average of enrollment samples)
- Matching: cosine similarity + margin check
- Scope: only students enrolled in the selected class

6.2 Attendance Rules

- Present mark is unique for (student, class, subject, date)
- Teachers only access assigned class–subject pairs
- Classroom mode shows **Review** labels (Auto-OK / Needs review) instead of raw confidence percentages
- Teacher confirms before final classroom save

6.3 AI Attendance Copilot (Rules-based)

The copilot answers natural questions from live database data, for example: - “Who is absent today in CSE-A / DBMS?”

- “Flag students with less than 75% attendance”
- “Summarize today’s classroom photo session”

This demonstrates an AI assistant layer without depending on an external paid LLM for the first version.

7. System Modules

Module	Description
Authentication & RBAC	Admin, Teacher, Student roles with secure hashed passwords
Admin Panel	Classes, subjects, teachers, assignments
Student Management	Add / edit / delete students, class enrollment
Face Enrollment	Guided multi-frame capture + quality checks
Model Training	Build embeddings, store centroids, prune raw captures
Live Marking	Real-time camera recognition
Classroom Marking	Multi-face detection from group photo
Review Workflow	Teacher validation of weak matches
Records & Analytics	History, filters, percentage reports, CSV
AI Copilot	Rules-based Q&A on attendance data
Database Layer	PostgreSQL permanent backend

8. Technology Stack

Layer	Technology
Frontend	HTML, CSS, JavaScript (camera APIs)
Backend	Python, Flask
Face AI	InsightFace, OpenCV, NumPy

Layer	Technology
Matching	Cosine similarity on embeddings
Database	PostgreSQL (SQLite only as emergency fallback)
Security	Password hashing, sessions, role checks, rate limits
Deployment readiness	Gunicorn / Docker notes for production

9. Hardware / Software Requirements

Software: Python 3, Flask, InsightFace/ONNX Runtime, PostgreSQL (Postgres.app or server), modern browser with camera access.

Hardware (minimum demo): Laptop/PC with webcam, 8 GB RAM recommended for face model. For classroom photos, a good phone/camera image is enough.

10. Advantages

- Reduces proxy attendance compared with manual roll call
 - Saves classroom time
 - Supports both individual and group photo marking
 - College-aware design (class/subject/teacher rights)
 - Scalable storage using embeddings instead of keeping dozens of photos forever
 - Teacher-in-the-loop review improves reliability
 - AI Copilot makes the system interactive and demo-friendly
 - PostgreSQL avoids data split across SQLite and Postgres
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11. Limitations

- Recognition depends on lighting, pose, occlusion, and camera quality
 - Far seats in large classrooms are harder to match
 - Current copilot is rules-based (not a full generative LLM)
 - Needs good enrollment quality and occasional re-capture if appearance changes a lot
 - Server setup required for multi-teacher concurrent use
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12. Future Scope

1. Liveness detection (anti-spoofing)
 2. Multi-frame confirmation before live save
 3. Dedicated face-recognition worker process
 4. FAISS / vector index for 7000+ students
 5. LLM-powered copilot with grounded DB tools
 6. College portal API sync
 7. Mobile app for teachers
 8. Audit logs, CSRF, HTTPS hardening for full production
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13. Conclusion

The Smart Digital Attendance System combines **computer vision (InsightFace)** with a **practical college web workflow**. It automates attendance using face embeddings, supports live and classroom modes, keeps teachers in control through review, stores data reliably in PostgreSQL, and adds an AI Copilot for academic queries.

The project is suitable as a major / course project because it demonstrates:

- Real AI usage (face embeddings + matching)
- Software engineering (roles, modules, database design)
- Product thinking (review workflow, storage scale, copilot)

It is not only a face-demo; it is an **AI-assisted institutional attendance platform** designed for classroom reality.

14. References (indicative)

1. InsightFace documentation / ArcFace face recognition literature
 2. Flask documentation
 3. PostgreSQL documentation
 4. OpenCV image processing references
 5. Related academic reports on biometric attendance systems
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End of Synopsis